

AI/ML PROJECT SUBMISSION



PROJECT OVERVIEW	STUDENT DETAILS
• Project Number: 2 • Task Name: Library Management System. • Difficulty: Easy to Moderate	• Name: Pranav Gajanan Rane • College: Government Polytechnic Gondia • Branch: Computer Engineering • Domain: AI/ML



```
from datetime import date,timedelta
```

```
def add_book() : #Add book in collection
```

```
    try :
```

```
        isbn = input("Enter the ISBN of the book - ")
```

```
        if isbn in library :
```

```
            print("A book with this ISBN already exists!")
```

```
        else :
```

```
            title = input("Enter the title of the book - ")
```

```
            author = input("Enter the authors name - ")
```

```
            print("Book added successfully!")
```

```
            library[isbn] = {
```

```
                "title" : title,
```

```
                "author" : author,
```

```
                "available" : True,
```

```
                "borrower_name" : None,
```

```
                "borrower_id" : None,
```

```
                "issue_date" : None,
```

```
                "due_date" : None
```

```
            }
```

```
    except Exception:
```

```
        print("Something Unexpected happened!")
```

```
def issue_book() : # Issue book
```

```
    try :
```

```
        issue = input("Enter the ISBN to get the book!")
```

```
        if issue in library :
```

```
            if library[issue]['available'] == True :
```



```
print("The book is available to be issued!")
library[issue]['available'] = False
    name = input("Enter your name - ")
    library_id = int(input("Enter your library id - "))
    library[issue]['borrower_name'] = name
    library[issue]['borrower_id'] = library_id
    today = date.today()
    library[issue]['issue_date'] = today
    due_date = today + timedelta(days=7)
        print(f"You need to return the book by -
{due_date}")
        print(f"You, {library[issue]['borrower_name']}
have successfully borrowed the book!")
        library[issue]["due_date"] = due_date

else :
    print("The book is unavailable to be borrowed! ")

else :
    print("The book with this ISBN isn't available in our
library!")
except ValueError :
    print("Pls enter correct value!")

def return_book() : #Returns a book
    try :
        issue = input("Enter the ISBN of book to be returned
- ")
```



if issue in library:

if library[issue]['available'] == True :

print("The book was not issued!")

else :

library[issue]['borrower_name'] = None

library[issue]['borrower_id'] = None

library[issue]['issue_date'] = None

library[issue]['due_date'] = None

library[issue]['available'] = True

print("The book was returned successfully!")

else:

print("The book doesn't belong to this library")

except ValueError:

print("Pls enter correct values!")

def view_catalog() : # Showcases all books(if available)

if not library:

print("No books to showcase!")

else :

fmt = "{:<20} {:<15} {:<15} {:<8}"

print("=" * 70)

**print(fmt.format("ISBN", "Title", "Author",
"Available"))**

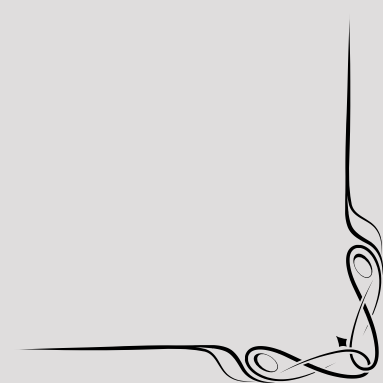
print("=" * 70)

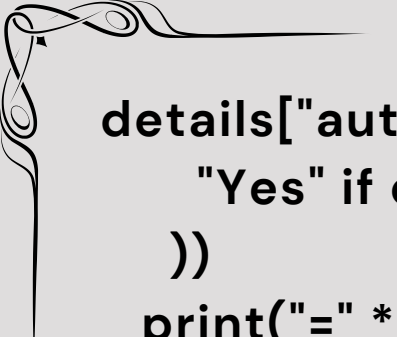
for key,details in library.items() :

print(fmt.format(

key,

details["title"],





```
details["author"],
    "Yes" if details["available"] else "No",
))
print("=" * 70)
```

```
def check_due_date() : # Checks due date using
issued date
    try :
        issue = input("Enter the ISBN of the book - ")
        if issue not in library:
            print("This book was not issued from our library!")
        else :
            if library[issue]['available'] == True :
                print("The Book was not issued!")
            else :
                print(f"The due date of your borrowed book is -
{library[issue]['due_date']}")
        except ValueError :
            print("Enter correct value!")
```

```
def search_book(): #Bonus Task -- Searches Books by
author or title
    try :
        if not library:
            print("No books to showcase!")
        else :
            choice = input("Do you want to search a specific
book or books issued by a author(type Title or Author)
- ").lower()
```



```
if choice == 'title' :
    title = input("Enter the title of the book - ").lower()
    found = False
    for isbn,details in library.items():
        if title == details['title'].lower() :
            found = True
            print("\n" + "=" * 50)
            print(f"  BOOK DETAILS FOUND (ISBN: {isbn})")
            print("=" * 50)
            print(f" Title:  {details['title']}")
            print(f" Author: {details['author']}")
            print(f" Available: {"YES" if details['available']}
else "NO"})
            print(f" Issued on : {details['issue_date']}")
            print(f" Due Date : {details['due_date']}")
            print(f"Borrower id(if borrowed) :
{details['borrower_id']}")
            print("=" * 40)
        if not found :
            print("No book found with such Title!")
    elif choice == 'author':
        name = input("Enter the Author's name - ").lower()
        found = False
        for isbn,details in library.items():
            if name == details['author'].lower() :
                found = True
print("\n" + "=" * 50)
        print(f"  FOUND AUTHOR'S BOOK'S  ")
        print("=" * 50)
```

```
print(f" Title:  {details['title']}")
    print(f" Available: {"YES" if {details['available']}
else "NO"}")
    print(f" Issued on : {details['issue_date']}")
    print(f" Due Date : {details['due_date']}")
        print(f"Borrower id(if borrowed) :
{details['borrower_id']}")
    print("=" * 40)
    if not found :
        print("No Author Books found!")
    else :
        print("Pls enter only author or title!")
except ValueError:
    print("Pls enter correct values!")
```

```
def show_menu() : #Shows what you can perform
    print("What you can perfrom - ")
    print("1)Register a new book")
    print("2)Issue a book to a student")
    print("3)Return a Book")
    print("4)View all books")
    print("5)Check the due date for a certain book")
    print("6)Search a book by its author or title.")
    print("7)Exit...")
```

""Actual System Below ""

```
library = {} # Will store all data
```

elif choice == "3" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]['Maths_Marks'] = mark1

print("Marks Updated!")

elif choice == "4" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]

['Social_Science_Marks'] = mark1

print("Marks Updated!")

elif choice == "5" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]['H/S_Marks'] = mark1

print("Marks Updated!")

elif choice == "6" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]['English_Marks'] =

mark1

print("Marks Updated!")

else :

print("Pls Enter the correct value(1-6)")

total = student_records[roll_no]['Science_Marks']

+ student_records[roll_no]['Social_Science_Marks'] +

student_records[roll_no]['Maths_Marks'] +

student_records[roll_no]['English_Marks'] +

student_records[roll_no]['H/S_Marks']

percentage, grade = calculate_grade(total)

student_records[roll_no]['percentage'] =

percentage

student_records[roll_no]['grade'] = grade

Note - This code might break due to incorrect indentation while pdf making so pls use the code mentioned in .py file attached

TEST OUTPUTS -

```
C:\Users\prana\OneDrive\Desktop\Python_Mini_Projects\Project2_Library_Management>C
neDrive/Desktop/Python_Mini_Projects/Project2_Library_Management/library_management

-----
_____Welcome to Library Management System!_____
Note - Asks for user input to perform a specific action.
-----

What you can perfrom -
1)Register a new book
2)Issue a book to a student
3)Return a Book
4)View all books
5)Check the due date for a certain book
6)Search a book by its author or title.
7)Exit...
Enter what you want to do (1 - 7) - 1
-----

Enter the ISBN of the book - 101
Enter the title of the book - Mahabharat
Enter the authors name - Ved Vyas
Book added successfully!
```

```
_____Welcome to Library Management System!_____
Note - Asks for user input to perform a specific action.
-----

What you can perfrom -
1)Register a new book
2)Issue a book to a student
3)Return a Book
4)View all books
5)Check the due date for a certain book
6)Search a book by its author or title.
7)Exit...
Enter what you want to do (1 - 7) - 4
-----

=====
ISBN                Title                Author                Available
=====
101                 Mahabharat          Ved Vyas              Yes
102                 The wonders         JK Rowling            Yes
Ramayan             Ramayan             Ved Vyas              Yes
106                 Maths for 10th      Nilam Production      Yes
=====
```

```
-----
_____Welcome to Library Management System!_____
Note - Asks for user input to perform a specific action.
-----
```

```
What you can perfrom -
1)Register a new book
2)Issue a book to a student
3)Return a Book
4)View all books
5)Check the due date for a certain book
6)Search a book by its author or title.
7)Exit...
Enter what you want to do (1 - 7) - 2
-----
Enter the ISBN to get the book!101
The book is available to be issued!
Enter your name - Pranav
Enter your library id - 95454
You need to return the book by - 2026-06-29
You, Pranav have successfully borrowed the book!
```

```
1)Register a new book
2)Issue a book to a student
3)Return a Book
4)View all books
5)Check the due date for a certain book
6)Search a book by its author or title.
7)Exit...
Enter what you want to do (1 - 7) - 4
-----
```

```
=====
ISBN          Title          Author        Available
=====
101           Mahabharat   Ved Vyas      No
102           The wonders JK Rowling    Yes
Ramayan       Ramayan     Ved Vyas      Yes
106           Maths for 10th Nilam Production Yes
=====
```

```
-----
_____Welcome to Library Management System!_____
Note - Asks for user input to perform a specific action.
-----
```

```
What you can perfrom -
1)Register a new book
2)Issue a book to a student
3)Return a Book
4)View all books
5)Check the due date for a certain book
6)Search a book by its author or title.
7)Exit...
Enter what you want to do (1 - 7) - 5
-----
Enter the ISBN of the book - 101
The due date of your borrowed book is - 2026-06-29
```

```
Enter what you want to do (1 - 7) - 6
-----
Do you want to search a specific book or books issued by a author(type Title or Author) - author
Enter the Author's name - ved vyas

=====
                FOUND AUTHOR'S BOOK'S
=====
Title:           Mahabharat
Available:  YES
Issued on :   2026-06-22
Due Date :     2026-06-29
Borrower id(if borrowed) : 95454
=====

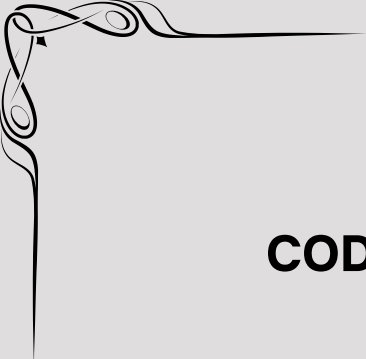
=====
                FOUND AUTHOR'S BOOK'S
=====
Title:           Ramayan
Available:  YES
Issued on :   None
Due Date :     None
Borrower id(if borrowed) : None
```

TEST OUTPUTS IN BROKEN CASES -

```
Enter what you want to do (1 - 7) - 6
-----
Do you want to search a specific book or books issued by a author(type Title or Author) - uu
Pls enter only author or title!
```

```
Enter what you want to do (1 - 7) - 1
-----
Enter the ISBN of the book - 101
A book with this ISBN already exists!
```

```
Enter what you want to do (1 - 7) - 2
-----
Enter the ISBN to get the book!101
The book is unavailable to be borrowed!
```



CODE TESTING WAS DONE SUCCESSFULLY!

More Details about the project is mentioned in the .md file attached . This pdf was made specifically to attach the code and its outputs.

PROJECT COMPLETED

